mmml — zsh

MMML

end-to-end MLIP toolkit

// data · train · MD · spectra · deploy · 31+ CLI commands

$ mmml —help

// COSMO Lab · EPFL

mmml — zsh

EricBoittier/mmml

docs

EricBoittier/mmml_tutorial

CI

// contributions from Sena Aydin, Khamlek Chaton, and everyone who has built MMML

MLIP workflow — generate · train · simulate · analyze

mmml — zsh

GPU reference data

```
mmml pyscf-dft
mmml pyscf-mp2
mmml pyscf-evaluate
mmml normal-mode-sample
mmml xml2npz
mmml fix-and-split
mmml validate
mmml active-learning
```

train & evaluate

```
mmml physnet-train
mmml train-joint
mmml ef-train
mmml physnet-evaluate
```

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MD & hybrid MM/ML

```
mmml md-system
mmml physnet-md
mmml ef-md
// CHARMM MLpot · JAX-MD · λ-dynamics TI
```

spectra & conformers

```
mmml-spectra-md
// IR · VCD · Raman
mmml orca-server
// ExtOpt GOAT
mmml gui
mmml lambda-mbar
```

Hybrid CHARMM integration is one deployment path — not the whole repo.

Atomistic ML software — MMML today, Metatomic next?

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CI/CD · tests

✓ python -m build pass

✓ pytest tests/ pass

✓ mkdocs build --strict pass

65+ unit tests · pycharmm | gpu | mlpot

// Python 3.13 · uv · JAX JIT · Pydantic · NPZ schema · Ruff · mypy

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// metatensor/metatomic · ORCA ExtOpt

issue 228

PR metatensor/metatomic#262 +1328 ● open

PR 262: metatomic-orca-server / client

MMML: mmml orca-server · GPU batch · FastAPI

Same ExtOpt protocol in JAX (MMML) and metatomic-torch → COSMO